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JD521 Formative Assessment 1

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Section A: Multiple Choice Questions

1. Which of the following is a valid declaration of a char array in Java?

   C. char[] chars = new char[]{'J', 'a', 'v', 'a'};

2. What is the result of the following code snippet?

   C. 2.0

3. Which of the following for-loop declarations is invalid in Java?

   D. for(int i = 0, String s = "Java"; i < s.length(); i++){}

4. What will the following switch statement print?

   B. TwoThreeDefault

5. Which principle of OOP is violated when a subclass changes the behaviour of a superclass method?

   C. Polymorphism.

6. What is the output of the following code snippet, assuming all necessary imports are done and files exist?

   B. true false

7. Which of the following is true about the finally block in Java?

   C. It is executed even if there is a return statement in the try or catch block.

8. Consider the following Code snippet. What will the output be?
   C. Arithmetic Exception Finally block

9. Which of these is not a difference between HashSet and TreeSet in Java?

   D. HashSet is faster for operations like add, remove, and contain compared to TreeSet.

10. What will happen if you try to compile and run the following code snippet?
    A. Compilation error at Line X

11. Which tool is used for static code analysis in Java to find bugs, vulnerabilities, and code smells?

    C. SonarQube

12. What is the purpose of the JVM (Java Virtual Machine)?

    B. To provide a runtime environment for executing Java bytecode

13. Which keyword is used to manually throw an exception in Java?
    A. throw

14. When is a try-with-resources statement useful in Java?
    B. When the resources used, like files or sockets, implement the AutoCloseable or Closeable interface and should be closed automatically after use.

15. What is true about the Comparator interface in Java?
    C. It is found in the java.util.Comparator package.

16. Which of the following statements about generics in Java is incorrect?
    D. Generics can only be used with collection classes like ArrayList and HashMap.

17. What is the function of the JDK (Java Development kit)?
    C. It includes tools for developing, debugging, and monitoring Java applications

18. Which of the following tools is used to manage dependencies and build Java projects?

    C. Maven

19. In Java, which of the following is not a valid way to create a thread?
    D. Directly instantiating the Thread class with a lambda expression

20. Which statement about lambda expressions in Java is true?
    B. They allow the function to be treated as an argument, or code as data.

Section B: Scenario Questions

```
1  import java.utl.Scanner; Import statement error
2
3  public classe DebuggingExercise {'Class' spelling error
4
5      public static void Main(String[] args) {'Main' case/decleration errror
6          Scanner input = new Scanner(System.in);
7
8          Sstem.out.println("Enter an integer: "); 'System' spelling error
9          int number = input.nextint(); 'Int' method call error
10
11         int result = sumNumbers(upTo: 5); Syntax error
12         System.ot.println("Sum up to 5 is: " + result); Syntax error/ print statement error
13
14         if(number = 10) { Operation error
15             System.out.println("You entered ten!");
16         }
17
18         // Calculate factorial
19         int factorial = 1;
20         for(int i = 2; i <= number i++) {Syntax error
21             factorial *= i;
22         }
23
24         System.out.println("Factorial of " + number + "is: Print statement error
   " + factorial);
25
26                 int divisionResult = number / 0; Divison by 0
27
28         System.out.println("Division result: " +
   divisionResult);
29     }
30
31     pubic static int sumNumbers(int upTo) { 'Public' spelling error
32         int sum = 0;
33         for(int i = 0; i <= upTo; i++) Loop Condition error
34             sum += i;
35         return sum Missing semicolon
36     }
       Syntax error
```

Debugging Exercise Correct Answers

- import java.util.Scanner;
- public class DebuggingExercise {
- public static void main(String[] args) {
- System.out.println("Enter an integer: ");
- int number = input.nextInt();
- int result = sumNumbers(5);
- System.out.println("Sum up to 5 is: " + result);
- if(number == 10) {
- for(int i = 2; i <= number; i++) {
- Division by zero is not corrected as it's mathematically incorrect. I'll add a comment to inform about this issue.
- public static int sumNumbers(int upTo) {
- for(int i = 1; i <= upTo; i++)
- return sum;
- System.out.println("Factorial of " + number + " is: " + factorial);

Section C: Pseudocode

Import the InputMismatchException and Scanner classes from the java.util package.

Create a class called Calculator to perform arithmetic operations.

Define methods in the Calculator class for addition, subtraction, multiplication, and division.

In the divide method, throw an ArithmeticException if the first number is zero to prevent division by zero.

Create a class called mainCalculator containing the main method.

Inside the main method:

- Create a Scanner object to read input from the user.
- Create a Calculator object to perform operations.
- Use a try-catch block to handle exceptions that may occur during input.
- Prompt the user to enter the first number.
- Prompt the user to select an arithmetic operator (+, -, *, /).
- Prompt the user to enter the second number.
- Use a switch statement to perform the selected operation.
- Display the result of the operation to the user.
- If the input is not a valid numeric value, catch the InputMismatchException and display an error message.
- If there's a division by zero, catch the ArithmeticException and display a message.

Close the Scanner object.

Use cases

Basic Arithmetic Operations:

- Users can perform basic arithmetic operations such as addition, subtraction, multiplication, and division on two numbers.
- Use Case: A person wants to quickly calculate the result of adding two numbers to calculate a mathematic equation.

Error Handling:

- The program provides user-friendly error messages when incorrect input is entered, the program displays understandable error messages to enhance user experience and assist users in fixing their errors.
- Use Case: When a user uses the calculator and unintentionally types a letter rather than a number, the application prompts them to enter a valid number.

Educational Tool:

- Students learning fundamental arithmetic operations can benefit from using the calculator as an instructional tool.
- Use Case: To teach primary school pupils about addition and subtraction, a teacher also utilizes a calculator in the classroom.

Input Output Process

Input:

- Two numbers and a arithmetic operations (addition, subtraction, multiplication, division).
- User input for selecting the arithmetic function to operate.

Process:

- Read input from the user for two numbers.
- Read the user's input for selecting the arithmetic function.
- Perform the selected arithmetic operation for the two numbers selected by the user.

Output:

- Display the output of the arithmetic operation to the user.
- Display a friendly error message if:
    - The input provided is not a valid numeric value (InputMismatchException).
    - The division by zero is used during division operation (ArithmeticException)